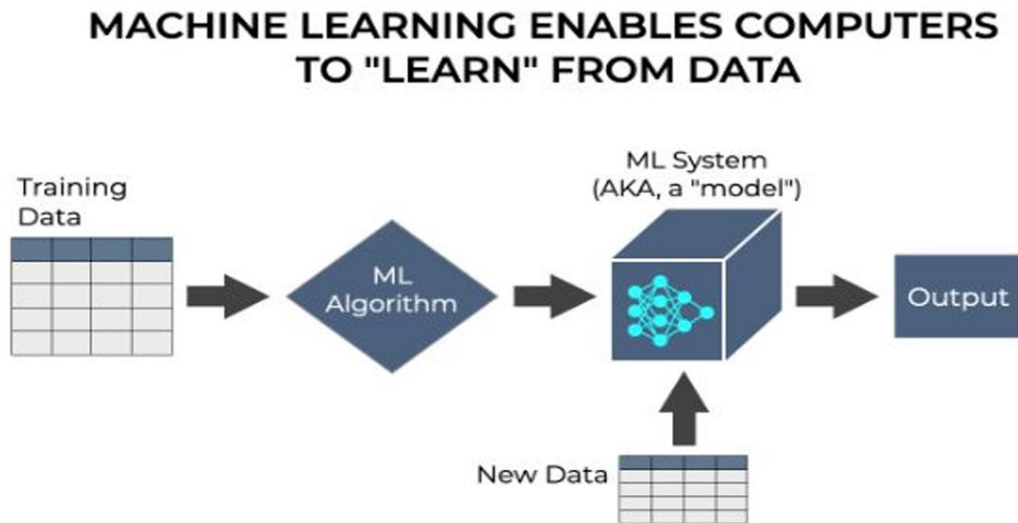


ML IAE I

1. Define ml and explain with example importance of ml



- Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed.
- ML is a branch of AI where a computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks T, as measured by P, improves with experience E.

- **Examples**

- **Handwriting recognition learning problem**

- Task T : Recognizing and classifying handwritten words within images
- Performance P : Percent of words correctly classified
- Training experience E : A dataset of handwritten words with given classifications

- **A robot driving learning problem**

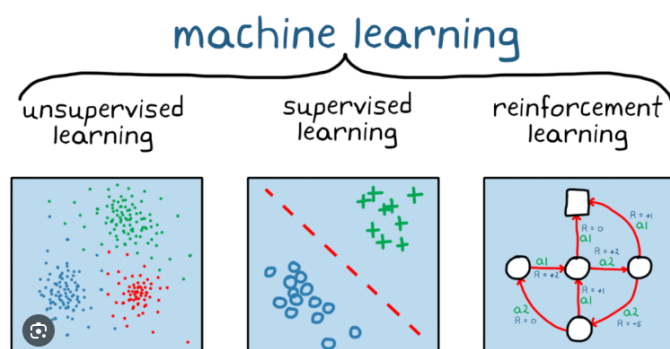
- Task T : Driving on highways using vision sensors
- Performance P : Average distance traveled before an error
- Training experience E : A sequence of images and steering commands recorded while observing a human driver

Importance of Machine Learning:

1. **Automation** – ML enables systems to make decisions without explicit programming.
2. **Handling Big Data** – Efficiently processes **large-scale, high-dimensional, unstructured data** (images, text, audio).
3. **Improves with Experience** – Performance **enhances automatically** with more data and time.
4. **Wide Applications** – Used in **healthcare (disease prediction), finance (fraud detection), e-commerce (recommendations), automobiles (self-driving cars), natural language processing (chatbots), image/speech recognition**, etc.
5. **Business Growth** – Companies like **Amazon, Google, Netflix** rely on ML for **personalized recommendations and user experience optimization**.

2. Explain how supervise learning is different from unsupervised learning

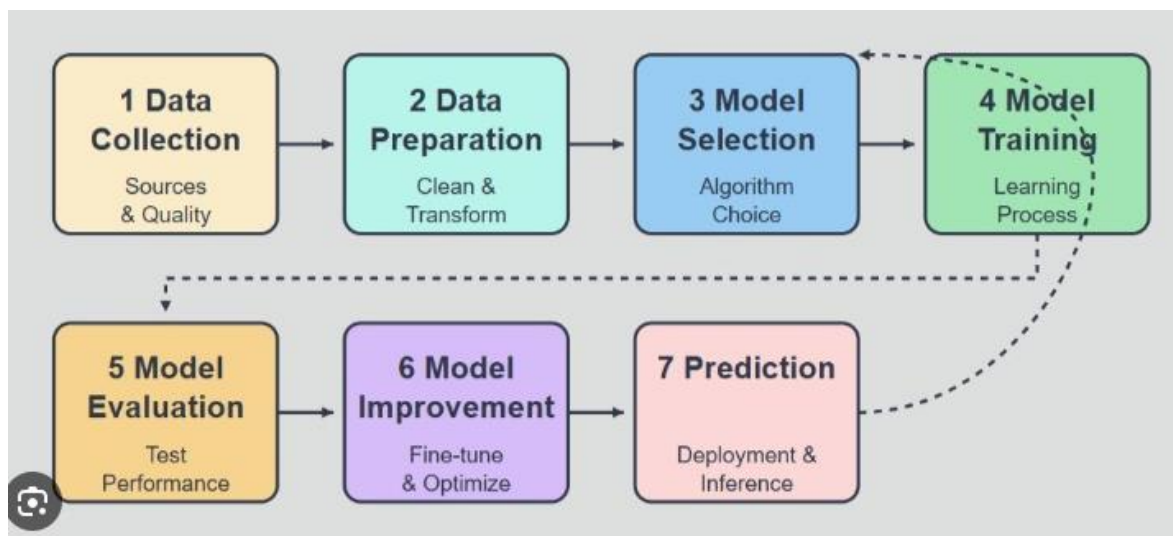
Supervised Learning	Unsupervised Learning
Supervised learning algorithms are trained using labeled data.	Unsupervised learning algorithms are trained using unlabeled data.
Supervised learning model takes direct feedback to check if it is predicting correct output or not.	Unsupervised learning model does not take any feedback.
Supervised learning model predicts the output.	Unsupervised learning model finds the hidden patterns in data.
In supervised learning, input data is provided to the model along with the output.	In unsupervised learning, only input data is provided to the model.
The goal of supervised learning is to train the model so that it can predict the output when it is given new data.	The goal of unsupervised learning is to find the hidden patterns and useful insights from the unknown dataset.
Supervised learning needs supervision to train the model.	Unsupervised learning does not need any supervision to train the model.
Supervised learning can be categorized in Classification and Regression problems.	Unsupervised Learning can be classified in Clustering and Associations problems.
Supervised learning can be used for those cases where we know the input as well as corresponding outputs.	Unsupervised learning can be used for those cases where we have only input data and no corresponding output data.
Supervised learning model produces an accurate result.	Unsupervised learning model may give less accurate result as compared to supervised learning.
Supervised learning is not close to true Artificial intelligence as in this, we first train the model for each data, and then only it can predict the correct output.	Unsupervised learning is more close to the true Artificial Intelligence as it learns similarly as a child learns daily routine things by his experiences.
It includes various algorithms such as Linear Regression, Logistic Regression, Support Vector Machine, Multi-class Classification, Decision tree, Bayesian Logic, etc.	It includes various algorithms such as Clustering, KNN, and Apriori algorithm.



3. Explain the steps in developing ml application

The process of building a **Machine Learning application** can be broken down into **7 major steps**.

1. Gathering Data	Collect sufficient and diverse data with relevant features (e.g., color and shape) to avoid bias.
2. Preparing the Data	Clean, randomize, and split the dataset into training (~80%) and testing (~20%) for fair evaluation.
3. Choosing a Model	Select an appropriate ML algorithm (Supervised, Unsupervised, or Reinforcement) based on the problem.
4. Training the Model	Feed training data into the algorithm so it can learn patterns by adjusting weights and biases.
5. Evaluation	Test the model on unseen data using metrics like Accuracy, Precision, Recall, or RMSE to measure performance.
6. Hyperparameter Tuning	Improve the model by adjusting parameters such as learning rate, iterations, or initialization values.
7. Prediction (Deployment)	Deploy the trained model to make real-world predictions on new incoming data.



4. Explain training and generalization error

Training Error and Generalization Error

- **Training Error:**

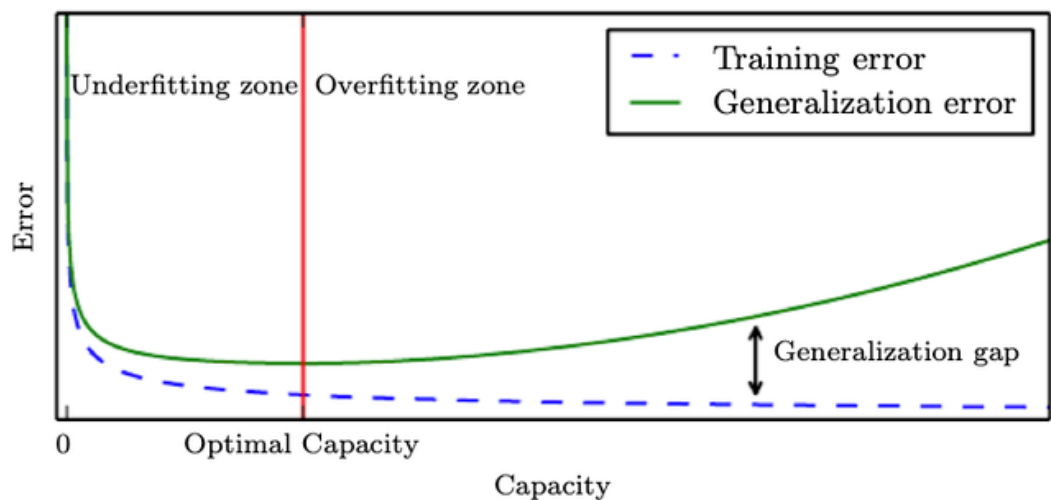
It is the error calculated on the **training dataset** that the model has already seen.

A low training error means the model has learned the given data well, but it may also indicate **overfitting** if the error is very low.

- **Generalization Error:**

It is the error measured on **unseen/test data** that was not used during training.

A low generalization error shows that the model can **generalize knowledge** and perform well in real-world situations.



5. Explain logistic regression

- **Definition:**

Logistic Regression is a **supervised learning algorithm** used for **classification problems**.

It predicts the **probability** of an outcome (Yes/No, 0/1, Spam/Not Spam) using the **logistic (sigmoid) function**.

- **Why "Regression"?**

Although used for **classification**, it is called regression because it uses a **linear regression-like approach** but applies a **sigmoid function** to map predictions into probability values between 0 and 1.

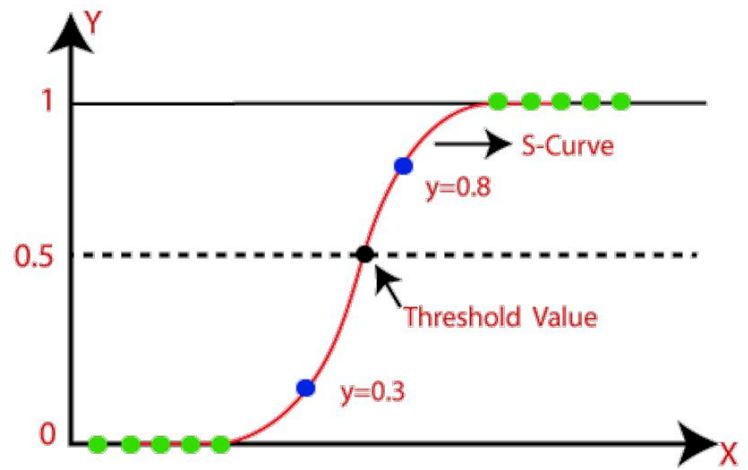
- **Key Idea:**

The model estimates:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

where

- β_0 = intercept,
- β_1 = coefficient (weight),
- e = base of natural log.
- **Decision Rule:**
If probability $\geq 0.5 \rightarrow$ Class 1 (e.g., Spam)
If probability $< 0.5 \rightarrow$ Class 0 (e.g., Not Spam)
- **Applications:**
 - Email Spam Detection (Spam/Not Spam)
 - Medical Diagnosis (Disease/No Disease)
 - Customer Churn Prediction (Leave/Stay)



Explanation of the Image (Sigmoid Curve in Logistic Regression)

- The X-axis represents the input values (linear combination of features).
- The Y-axis represents the **predicted probability** (always between 0 and 1).
- The red S-shaped curve (**Sigmoid function**) maps input values into probability values.
- At the middle, the curve passes through 0.5 (**Threshold Value**):
 - If probability $\geq 0.5 \rightarrow$ predicted as **Class 1** (e.g., Spam, Disease = Yes).
 - If probability $< 0.5 \rightarrow$ predicted as **Class 0** (e.g., Not Spam, Disease = No).
- Example points are shown:
 - $y = 0.3 \rightarrow$ below threshold $\rightarrow$ classified as 0.
 - $y = 0.8 \rightarrow$ above threshold $\rightarrow$ classified as 1.
- The green dots represent the **final classification outputs (0 or 1)** after applying threshold.

✓ In simple words:

This diagram shows how **Logistic Regression** converts input values into a **probability curve** using the **Sigmoid function** and then applies a **0.5 threshold** to classify the data into two classes.

6. Explain underfitting, overfitting, bias and variance tradeoff

1. Underfitting

- Underfitting occurs when the **model is too simple** and fails to capture important patterns in the training data.
- It is caused by **high bias** (strong assumptions by the model).
- The model performs **poorly on both training and test data**.
- Example: Using a **straight line** to fit non-linear data.
- This situation shows that the model has **not learned properly** and cannot be used in production.

2. Overfitting

- Overfitting occurs when the **model is too complex** and learns both the actual patterns and the **noise** in the data.
- It is caused by **high variance** (too much sensitivity to training data).
- The model performs **very well on training data but poorly on unseen/test data**.
- Example: The model memorizes training images of cats and misclassifies a new image of a fox as a cat.
- Overfitting reduces the model's ability to **generalize** to new data.

3. Bias

- Bias is the **error due to overly simplistic assumptions** made by the learning algorithm.
- High bias leads to **missing important features** of the data.
- It results in **underfitting**, where the model cannot perform well on both training and test data.
- Bias reduces the model's ability to **learn patterns** effectively.

4. Variance

- Variance is the **error due to sensitivity to small changes/noise** in the training data.
- High variance makes the model **too closely fit** the training data.
- It results in **overfitting**, where the model performs well on training data but fails on new data.
- Variance reduces the model's ability to **generalize**.

5. Bias–Variance Tradeoff

- The Bias–Variance Tradeoff is the process of finding the **optimal balance** between bias and variance.
- If **bias is high**, the model underfits; if **variance is high**, the model overfits.
- The goal is to minimize the **total error = bias² + variance + irreducible error**.
- An optimized model should capture **essential patterns** (low bias) and also **generalize well** (low variance).
- This tradeoff ensures the model achieves **best accuracy on both training and unseen data**.

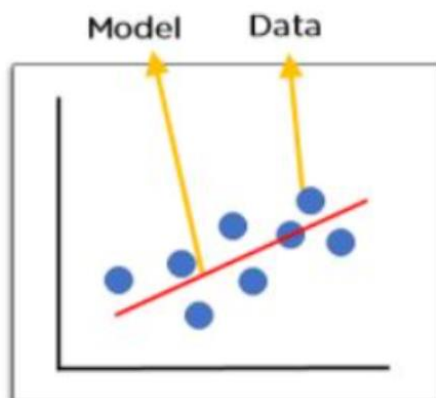


Figure 3: Underfitting

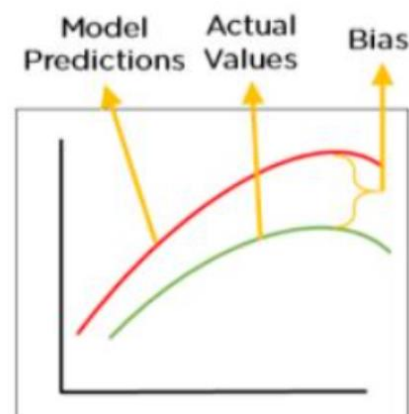
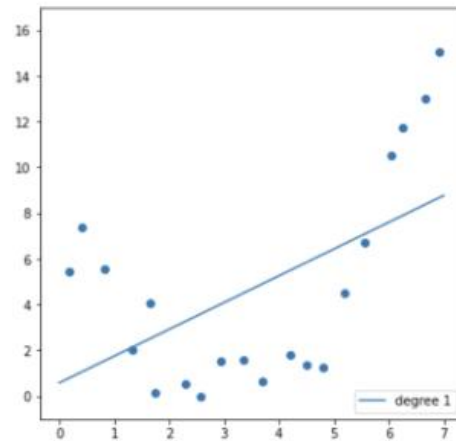
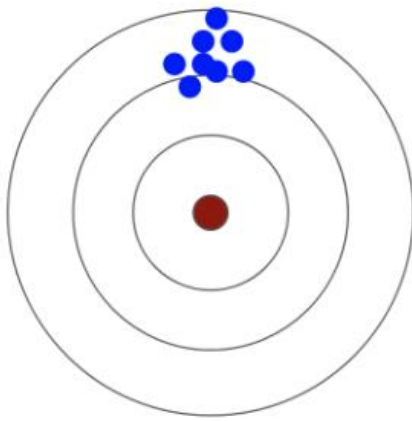


Figure 2: Bias

Low Variance and High Bias



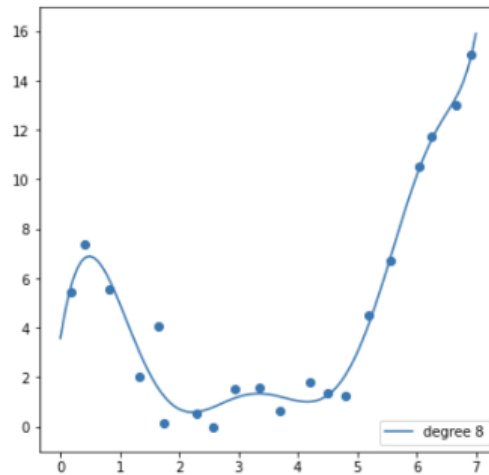
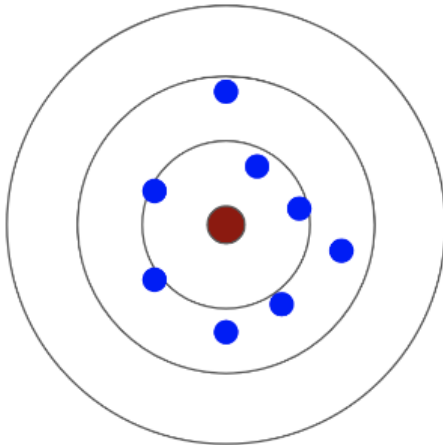
Underfitting



High Variance and Low Bias



Overfitting



7. Explain regression line, scatter plot, error in prediction and best fitting line

1. Regression Line

- A regression line is the line that represents the relationship between dependent and independent variables in regression analysis.
- It is expressed as:

$$Y = a + bX$$

where a = intercept, b = slope, X = independent variable, Y = dependent variable.

- The slope (b) shows the rate of change in Y for a unit change in X .
- The regression line is used for prediction of values.

2. Scatter Plot

- A scatter plot is a diagram that shows data points on a 2D graph with X-axis = independent variable, Y-axis = dependent variable.
- It is used to visualize the relationship between two variables.
- If points follow an upward trend → **positive correlation**; downward trend → **negative correlation**.
- The scatter plot helps us understand how closely the data follows a regression line.

3. Error in Prediction

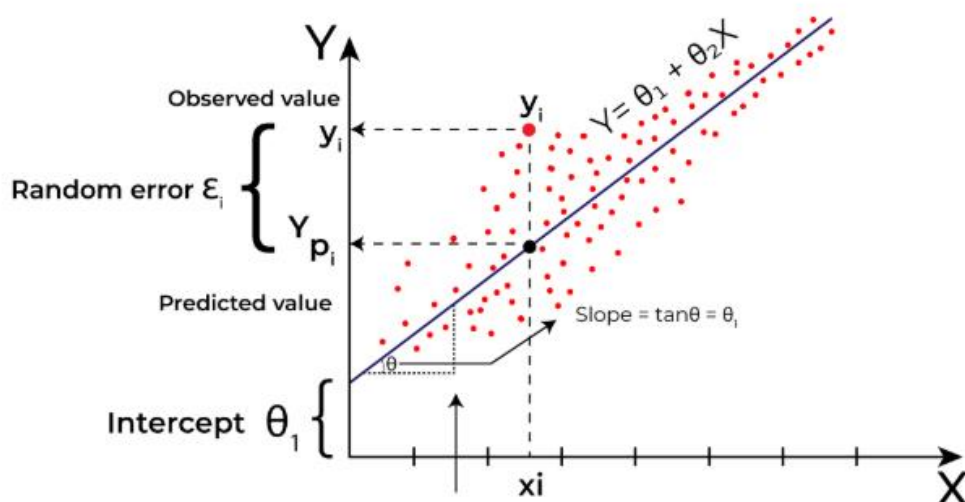
- Error in prediction is the difference between the actual value and the predicted value from the regression model.
- Mathematically:

$$Error = Y_{actual} - Y_{predicted}$$

- Errors indicate how well the regression line fits the data.
- Smaller errors → better accuracy; larger errors → poor fit.

4. Best Fitting Line

- The best fitting line is the regression line that minimizes the sum of squared errors (Least Squares Method).
- It passes through the data in such a way that the distance between actual data points and the line is **minimum**.
- The best fit line represents the optimal relationship between X and Y.
- It is widely used for prediction and forecasting.



8. Ensemble learning

Ensemble Learning

- **Concept:** Ensemble Learning is a powerful **machine learning technique** where **multiple models (learners)** are trained and combined to solve the same problem. Instead of relying on a **single weak model**, ensemble methods integrate the outputs of many models to produce a **strong, accurate, and generalized predictor**.
- **Key Idea:** “**Wisdom of the crowd**” – multiple weak learners (with individual errors) when combined can **cancel out each other’s weaknesses** and improve overall performance.
- **Goal:** To **reduce bias, variance, and prediction error** and achieve **better accuracy and robustness** than a single model.

Types of Ensemble Methods

1. Bagging (Bootstrap Aggregating)

- Uses **sampling with replacement** to create multiple training subsets.
- Trains models in **parallel** on these subsets.
- Final prediction is obtained by **majority voting (classification)** or **averaging (regression)**.
- **Reduces variance** and avoids overfitting.
- **Example:** Random Forest.

2. Boosting

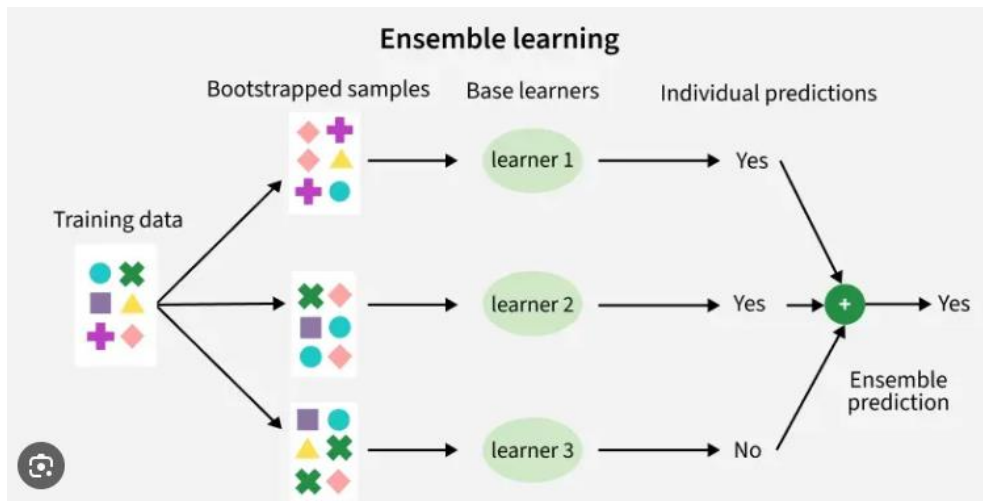
- Trains models **sequentially** where each new model focuses on correcting the **errors of the previous one**.
- Assigns **higher weights to misclassified samples**.
- Produces a strong learner by combining multiple weak learners.
- **Reduces bias** and improves accuracy.
- **Examples:** AdaBoost, Gradient Boosting, XGBoost.

3. Stacking (Stacked Generalization)

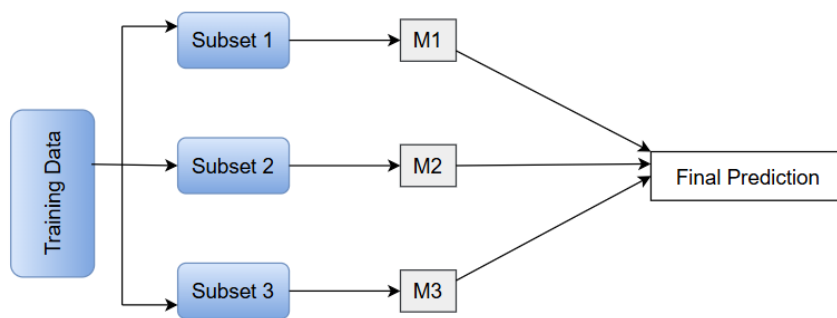
- Combines predictions of multiple base models using a **meta-model (higher-level model)**.
- Base models may be of **different types** (e.g., Decision Tree, SVM, Logistic Regression).
- The meta-model learns the best way to combine predictions.
- Provides **high flexibility and accuracy**.

Advantages of Ensemble Learning

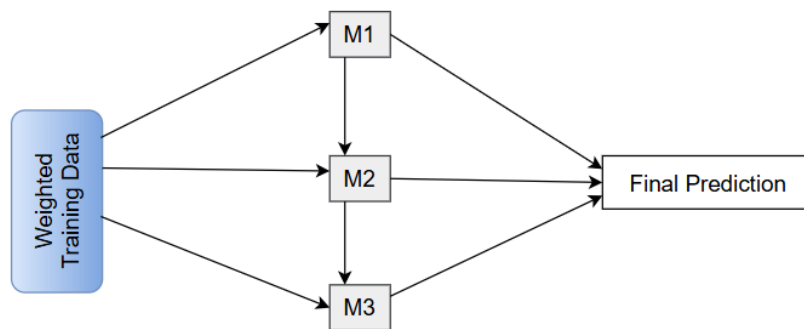
- Increases **accuracy and generalization**.
- Reduces **overfitting and variance**.
- Works well with **complex, noisy datasets**.
- Performs better in **real-world applications** like fraud detection, recommendation systems, medical diagnosis, and competitions like Kaggle.



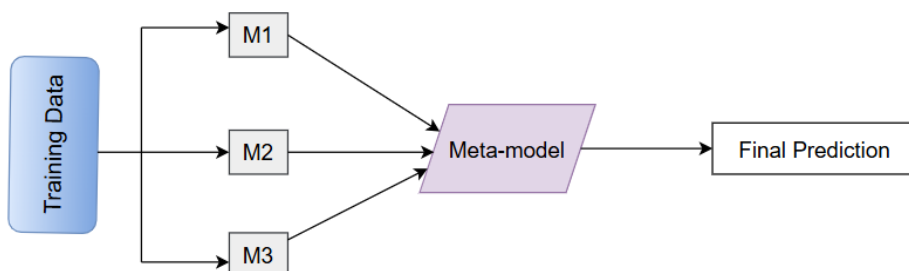
Bagging



Boosting



Stacking



9. Explain how regression problems can be solved using steepest descent method

Regression Problems using Steepest Descent Method

- **Concept:** The Steepest Descent Method (Gradient Descent) is an iterative optimization algorithm used to find the best-fitting regression line by minimizing the **cost function** (error).

- **Cost Function:**

In regression, we use the **Mean Squared Error (MSE)** as the cost function:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

where y_i = actual value, $\hat{y}_i$ = predicted value, m = number of samples.

- **Working of Steepest Descent:**

1. Initialize parameters ($\theta_0, \theta_1, \dots$) randomly.
2. Compute predictions using hypothesis:

$$\hat{y}_i = \theta_0 + \theta_1 x_i$$

3. Calculate error between predicted and actual values.
4. Update parameters in the opposite direction of gradient:

$$\theta_j := \theta_j - \alpha \frac{\partial J(\theta)}{\partial \theta_j}$$

where α = **learning rate** (step size).

5. Repeat steps until the cost function converges (minimum error).
- **Key Idea:** Move towards the **steepest downward slope** of the error function surface to gradually reach the **global minimum**.

- **Advantages:**

- Works well for **large datasets**.
- Provides an efficient way to compute the **best fit regression line**.
- Helps avoid manual trial-and-error in parameter selection.

10. Issues in Decision tree

1. **Overfitting**

- Decision trees may become too complex by fitting noise or irrelevant details in training data.
- This leads to **poor generalization** on unseen data.
- **Solution:** Pruning, limiting tree depth, or using ensemble methods.

2. **Biased towards features with more levels**

- Attributes with many distinct values (like ID numbers) may dominate the splitting process.
- This creates **biased splits** that do not improve accuracy.
- **Solution:** Use techniques like **Information Gain Ratio**.

3. **High Variance**

- Small changes in training data can produce **completely different tree structures**.
- This makes decision trees **unstable** and sensitive to data fluctuations.

4. Continuous Variables Handling

- Splitting continuous attributes requires finding **optimal thresholds**, which may increase complexity.

5. Computational Complexity

- Building a decision tree can be computationally expensive for **large datasets** because multiple splits must be evaluated.

6. Limited Expressiveness

- Decision trees work well for **axis-parallel splits**, but they may fail when relationships between attributes are complex or not linear.

11. Linear vs Logistic regression

Difference between Linear Regression and Logistic Regression:

Linear Regression	Logistic Regression
Linear regression is used to predict the continuous dependent variable using a given set of independent variables.	Logistic Regression is used to predict the categorical dependent variable using a given set of independent variables.
Linear Regression is used for solving Regression problem.	Logistic regression is used for solving Classification problems.
In Linear regression, we predict the value of continuous variables.	In logistic Regression, we predict the values of categorical variables.
In linear regression, we find the best fit line, by which we can easily predict the output.	In Logistic Regression, we find the S-curve by which we can classify the samples.
Least square estimation method is used for estimation of accuracy.	Maximum likelihood estimation method is used for estimation of accuracy.
The output for Linear Regression must be a continuous value, such as price, age, etc.	The output of Logistic Regression must be a Categorical value such as 0 or 1, Yes or No, etc.
In Linear regression, it is required that relationship between dependent variable and independent variable must be linear.	In Logistic regression, it is not required to have the linear relationship between the dependent and independent variable.
In linear regression, there may be collinearity between the independent variables.	In logistic regression, there should not be collinearity between the independent variable.